

Try LoRA on SmolVLA

Generate Dataset

**Try some pre-trained models
without fine-tuning**

Try LoRA on SmolVLA

How?

`pip install peft` (Parameter efficient fine-tuning with 🤗 PEFT)



```
1 policy:
2   type: smolvla
3   pretrained_path: lerobot/smolvla base
4   device: cuda
5   push_to_hub: false
6
7   # LoRA allows for a learning rate up to 10x higher compared to full-parameter fine-tuning.
8   optimizer_lr: 1e-3
9   scheduler_decay_lr: 1e-4
```

LoRA require a pre-trained base model for fine-tuning

LoRA allows for a learning rate up to 10x higher compared to full-parameter fine-tuning.



```
1 peft:
2   method_type: LORA
3   r: 64
```

enable LoRA

What are the results?

	smolvla_four_ merged	smolvla_four_ merged_lora
top long	68.33% 62.5%	31.67%
top short	11.67%	2.78%
pant long	38.33%	
pant short	85%	

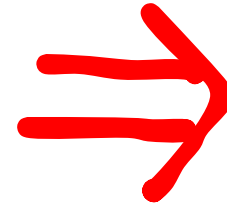
Generate Synthetic Data

**Really thanks for Charles
Sosmeña and Tan Thanh
Nguyen helping**

Implementation Steps

```
xvfb-run -a python -m scripts.dataset_sim replay \  
--dataset_root Datasets/example/record_pant_long_release_10/001 \  
--output_root Datasets/augmented/record_pant_long_release_10/001 \  
--task LeHome-BiSO101-Direct-Garment-v2 \  
--num_replays 1 \  
--device cpu \  
--enable_cameras \  
--start_episode 0 \  
--end_episode 2 \  
--save_successful_only
```

Make sure it can work



implement
dataset_replay_randomization.py



Create amounts of data

implement
dataset_merge.py

Combine four types of garment

Problem and Solving

scripts.dataset_sim replay

Problem

```
2026-04-03 09:19:57 - scripts.utils.dataset_replay - ERROR - Error during episode replay: Feature mismatch in `frame` dictionary:  
Extra features: {'observation.top_depth'}  
Traceback (most recent call last):  
  File "/home/mses101137/lifestone/lehome-challenge_S.N.N/scripts/utils/dataset_replay.py", line 446, in replay_episode  
    replay_dataset.add_frame(frame)  
  File "/home/mses101137/lifestone/lehome-challenge_S.N.N/.venv/lib/python3.11/site-packages/lerobot/datasets/lerobot_dataset.py", line 1140, in add_frame  
    validate_frame(frame, self.features)  
  File "/home/mses101137/lifestone/lehome-challenge_S.N.N/.venv/lib/python3.11/site-packages/lerobot/datasets/utils.py", line 1005, in validate_frame  
    raise ValueError(error_message)  
ValueError: Feature mismatch in `frame` dictionary:  
Extra features: {'observation.top_depth'}
```

Can't find feature of top_depth

Solution

hf download lehome/dataset_challenge --repo-type dataset --local-dir Datasets/example

dataset_merge

Problem

⋮

`FileNotFoundError: [Errno 2] No such file or directory: 'Datasets/augmented/record_pant_long_release_10/003/003/meta/tasks.parquet'`

⋮

`Repository Not Found for url: https://huggingface.co/api/datasets/ds_003/refs.`

`Please make sure you specified the correct `repo_id` and `repo_type`.`

`If you are trying to access a private or gated repo, make sure you are authenticated. For more details, see`

`https://huggingface.co/docs/huggingface\_hub/authentication`

`Invalid username or password.`

Some garment fail in every episode

Solution



failure case

Datasets/augmented/record_pant_long_release_10/003/003
Datasets/augmented/record_pant_long_release_10/009/009
Datasets/augmented/record_top_long_release_10/001/001
Datasets/augmented/record_top_long_release_10/007/007
Datasets/augmented/record_top_short_release_10/006/006
Datasets/augmented/record_top_short_release_10/010/010

dataset_merge

Problem

File "pyarrow/array.pxi", line 365, in pyarrow.lib.array

File "pyarrow/array.pxi", line 87, in pyarrow.lib._ndarray_to_array

File "pyarrow/array.pxi", line 75, in pyarrow.lib._ndarray_to_type

File "pyarrow/error.pxi", line 155, in pyarrow.lib.pyarrow_internal_check_status

File "pyarrow/error.pxi", line 92, in pyarrow.lib.check_status

pyarrow.lib.ArrowTypeError: ('Did not pass numpy.dtype object', 'Conversion failed for column observation.top_depth with type array[uint16]')

The observation.top_depth column contains nested uint16 arrays (a depth image stored as list<list<uint16>> in parquet). When the merge reads it via pd.read_parquet() and writes back via df.to_parquet(), the schema is lost and pyarrow can't auto-infer the nested type.

top_depth column can't maintain its schema during reading and writing

Solution

fix .venv/lib/python3.11/site-packages/lerobot/datasets/aggregate.py

✓ Datasets ●

✓ augmented ●

> four_types_merged ●

> record_pant_long_release_... ●

> record_pant_short_release... ●

> record_top_long_release_10 ●

> record_top_short_release_10 ●

> example ●

**Try some pre-trained models
without fine-tuning**

Why?

模型	top_long	top_short	pant_long	pant_short	
smolvla full FT <i>baseline</i>	68.33%	11.67%	38.33%	85.00%	<i>50.832</i>
smolvla pretrained <i>2.</i>	60.00%	30.00%	48.33%	83.33%	<i>55.415</i>
smolvla LoRA	31.67%	2.78%	—	—	
smolvla + prompt	0.00%	1.67%	—	—	
ACT	35.00%	—	—	—	

Smolvla pretrained: pretrained_path: lerobot/smolvla_base

Smolvla pretrained with : *3* 63.33 38.33 45 81.67 *57.082*

combined datasets

Smolvla libero_plus with : 60 28 32 85 *51.25*

combined datasets2

base

Smolvla pretrained with : combined datasets2	xx	35	31.67	81
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Smolvla pretrained with : combined datasets2	60	35	31.67	81
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